



Master's Degree Program in
Data Science and Management

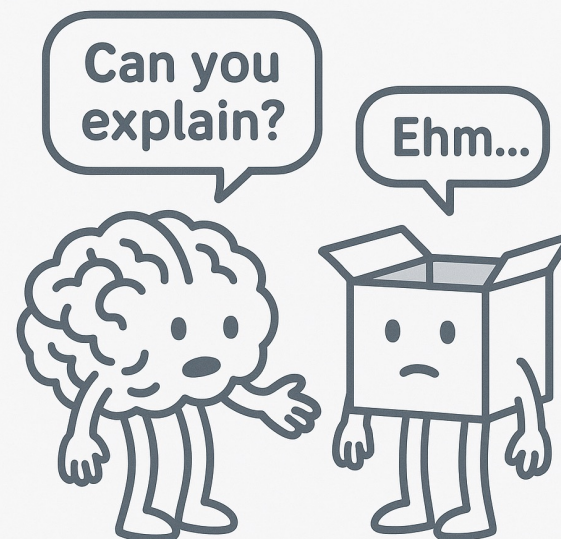
Course of Machine Learning

Supervisor
Prof. Giuseppe F. Italiano

Co-supervisor
Prof. Emilio Coppa

Candidate
Simone Luzi
782201

Explainable AI: Opening the Black Box for Reliable Decision Making



Why Explainable AI?



Every action we take arises from a reason we can make explicit

Most Machine Learning models function as black boxes



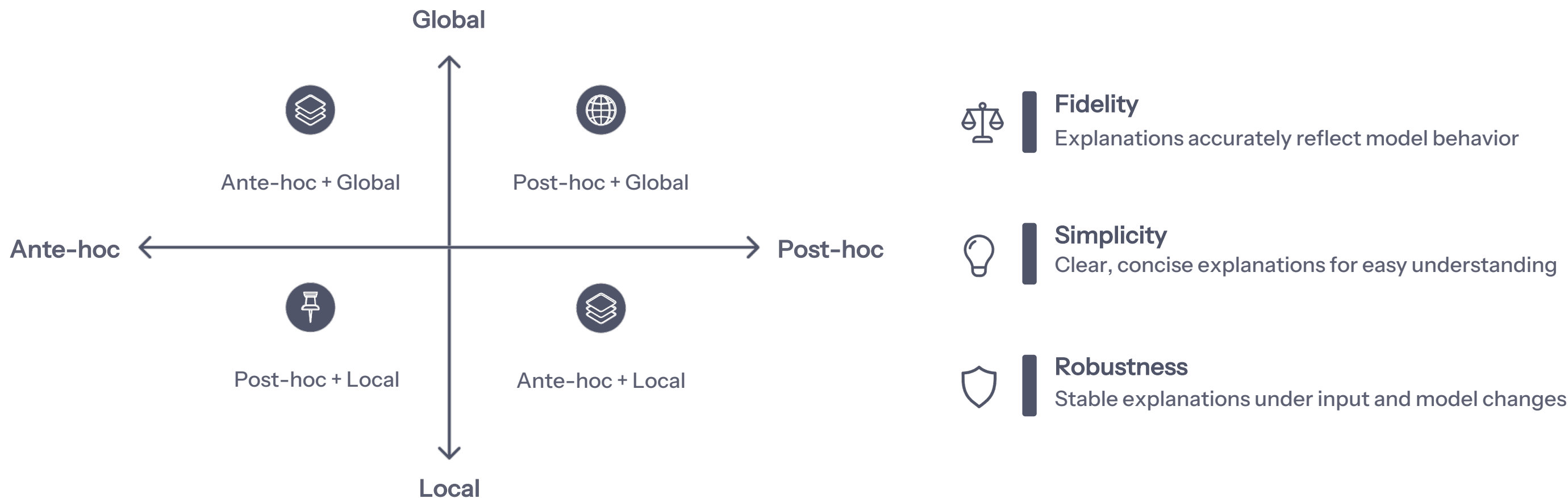
Application of Machine Learning algorithms in sensitive fields cannot afford opacity

Explainable AI brings the model's decision chain back into the open



Taxonomy of Explanations

The central element of XAI is explanations, organized along two fundamental axes that define their scope and application.



Explanations manifest as saliency maps, counterfactuals, textual rationales and attributional summaries.

Explainability Techniques

Traditional Approaches



Gradient-Based Methods

Link variations in the model's gradients to relevant parts of the input
I.E. GradCAM, Integrated Gradients



Perturbation-Based Methods

Measure how output changes occluding or replacing portions of the input
I.E. Sliding Window Occlusion, RISE



Model-Agnostic Approaches

Synthesise local contributions in a compact way.
LIME learns sparse linear surrogates
SHAP assigns additive shares with respect to a defined background

Advanced Techniques



Sparse Autoencoders

Learn a parsimonious and repeatable dictionary of concepts



Constrained Concept Refinement

Imposes simple and auditable constraints for signed, readable contributions



Mechanistic Interventions

Activation patching and ablation measure outcome shifts and distinguish correlations from causal levers



Microscopy

Map internal units and circuits through probes that check neural activation

Case Study: BreastMNIST Classification

Multilevel explainability pipeline applied to a ResNet-18_224 pre-trained on BreastMNIST.
BreastMNIST collects breast ultrasounds in B-mode in a compact single band format, with binary labeling.



Original Input

28×28 grayscale ultrasound images



Preprocessing

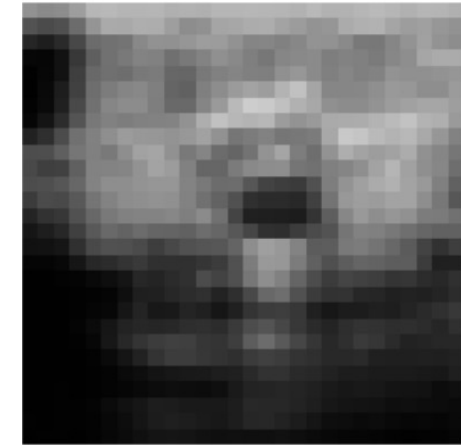
Resampled to 224×224 with bilinear interpolation, replicated across RGB channels



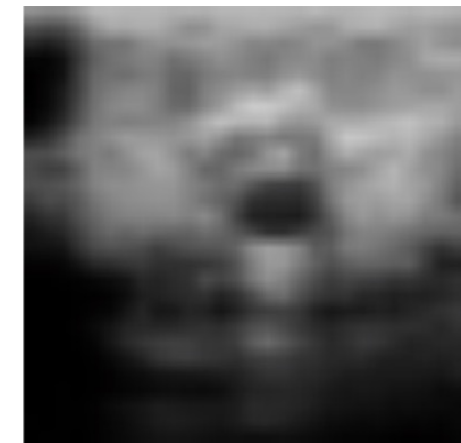
Normalization

Per-channel normalization with ImageNet or mean = std = 0.5

❏ The 7:1:2 dataset split for training, validation and test respectively follows the official scheme and is kept constant across different runs.



Original BreastMNIST 1×28×28



Preprocessed 3×224×224

Model and Re-heading Strategy

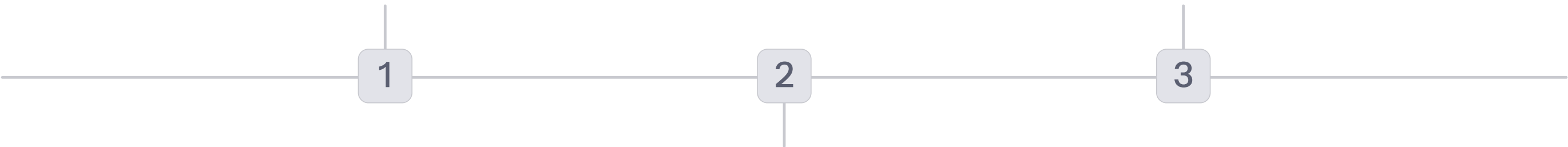
The neural network under examination belongs to the ResNet family, which uses residual connections to facilitate signal propagation adopted, loading the official weights and keeping the backbone frozen.

Original Configuration

ResNet-18_224 architecture pre-trained on BreastMNIST.
Official weights downloaded from Zenodo.
Backbone frozen to preserve powerful feature extraction capabilities.

Calibration

Temperature Scaling adjusts logits for calibrated probabilities.
Isotonic Regression remaps probabilities to correct remaining miscalibration.



Re-heading

Original classification head replaced with a single linear probe layer, trained on pooled features. Utilizes ℓ_2 regularization and class weights to prevent overfitting and address dataset imbalance.

Performance Improvements Across Seeds

The re-heading brings consistent improvement: more separation in the ranking, better binary decision, and probabilities that are on average more reliable.

Seed	Status	AUC	PR-AUC	ACC	BACC	Brier	ECE	AUC CI _{95%}
42	Pre-head	0.55	0.76	0.73	0.1	0.24	0.21	N/A
42	Re-head (baseline)	0.82	0.8	0.83	N/A	0.14	0.12	[0.73 – 0.90]
42	Re-heading + TS	0.82	0.89	0.83	N/A	0.14	0.11	[0.73 – 0.90]
42	Re-heading + IsoR	0.81	0.90	0.82	N/A	0.13	0.14	[0.72 – 0.89]
123	Pre-head	0.62	0.79	0.27	0.50	0.34	0.11	N/A
123	Re-head (baseline)	0.80	0.88	0.79	N/A	0.16	0.08	[0.72 – 0.88]
123	Re-heading + TS	0.80	0.88	0.79	N/A	0.15	0.11	[0.72 – 0.88]
123	Re-heading + IsoR	0.79	0.90	0.79	N/A	0.15	0.11	[0.71 – 0.86]
777	Pre-head	0.63	0.83	0.39	0.58	0.25	0.09	N/A
777	Re-head (baseline)	0.81	0.88	0.81	N/A	0.14	0.15	[0.71 – 0.88]
777	Re-heading + TS	0.81	0.88	0.81	N/A	0.14	0.16	[0.71 – 0.88]
777	Re-heading + IsoR	0.879	0.91	0.81	N/A	0.14	0.18	[0.71 – 0.87]

Local Explanations: Methods and Results

A panel of local explanations with gradient-based and agnostic methods is applied with a single setup:

- **Target:** logit of top 1 class;
- **Hook:** last convolutional block;
- **Pixel ordering:** absolute value of attribution;
- **Maps** computed on the uncalibrated head;
- **Quality** evaluated through deletion and insertion trajectories, and sensitivity- ϵ .

Integrated Gradients

Concentrates information early, with rapid collapse in deletion and fast reconstruction in insertion.

Grad-CAM++

Shows better local stability.

GradientSHAP

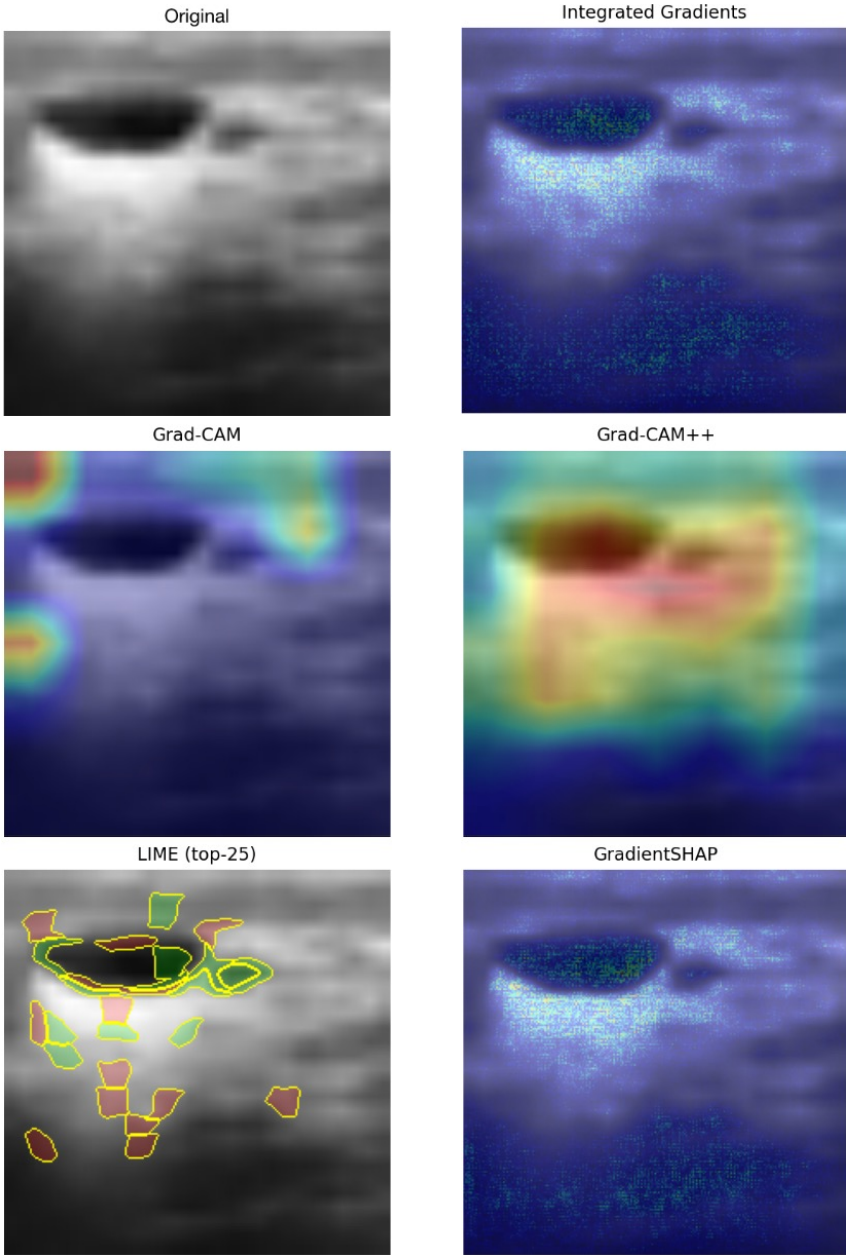
Concentrates attribution mass in the tail and centralizes energy over the field of view.

Grad-CAM

More aggressive in removal, slower reconstruction.

LIME with SLIC segmentation

Selects cohesive regions with limited activation and few dominant components.



Concept-Level Analysis: From Pixels to Interpretable Factors



Sparse Autoencoder

Trained on pooled features of frozen ResNet to obtain sparse, parsimonious and repeatable code where each unit corresponds to a recurring visual motif.

-
-
-

Atlas Stability

Shows stability and sparsity of the lexicon between training and test.



CCR-L Alignment

Constrained linear regression assigns signed weights to concepts and explains the dynamics of the logit.

-
-
-

CCR-L Performance

Achieves high adherence between concepts and logit with weights consistent with emerged semantics.



Micro-Interventions

Micro-patching in feature space measures logit shift when a single concept is forced in clean and corrupted runs.

-
-
-

Causal Validation

Micro-interventions order factors by impact and confirm effects in expected direction in the great majority of cases.

Future Developments

	Computational Optimization	Development Stabilize computational budget with regular measurements.	Benefit Better resource planning and regression detection.
	Cohort Expansion	Development Expand cohorts and replications on seeds and comparable versions.	Benefit Strengthen robustness of estimates.
	Threshold Definition	Development Define preregistered thresholds on faithfulness, stability and simplicity.	Benefit Increase causal validity with plausible baselines.
	Dictionary Refinement	Development Refine concept dictionary with simple and auditable constraints.	Benefit Improve readability and semantic alignment.
	Regulatory Alignment	Development Case reports with reason codes and calibrated confidences.	Benefit Multilevel explainability becomes reproducible evidence.

LUISS



**Thank You
for the attention!**